

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	CSE-AI	Date:	8-8-2026

Code of Conduct

/ P. KALYANI (192472199) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **P.Kalyani**

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

1. MLE of Bigram Probability $P(\text{science}|\text{data})$

The Maximum Likelihood Estimate for a bigram is:

$$P(w_i|w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$$

Given:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$

Therefore,

$$P(\text{science}|\text{data}) = \frac{C(\text{data science})}{C(\text{data})} = \frac{3}{3} = 1$$

Answer:

$$P(\text{science}|\text{data}) = 1.0$$

Interpretation:

The probability is **100%**, meaning every occurrence of "data" in the training corpus is followed by "science". Therefore, when a user types "**data**", the keyboard has very high confidence that the next word should be "**science**".

2. Backoff Model for the unseen sequence "data science improves"

The sequence "**data science improves**" is unseen because:

$$C(\text{data science improves})=0$$

A backoff model uses lower-order n-grams when a higher-order n-gram is unavailable.

For the trigram:

$$P(\text{improves}|\text{data,science})$$

there is no occurrence, so we back off to:

$$P(\text{improves}|\text{science})$$

But "**science improves**" is also not present in the given corpus.

Therefore, we back off further to the unigram:

$$P(\text{improves})$$

The word "**improves**" does not occur in the corpus, so:

$$P(\text{improves})=0$$

Thus, using only the supplied corpus statistics:

$$P(\text{data science improves})=0$$

Why backoff is essential:

Real-time keyboards constantly encounter new words and word combinations. Backoff prevents the system from depending only on exact trigram matches. It can use bigram or unigram information when a higher-order sequence is unseen. In a real system, smoothing is normally combined with backoff so that unseen words do not automatically receive zero probability.

3. Deleted Interpolation for "data science is"

Deleted interpolation combines trigram, bigram, and unigram probabilities:

$$P(w_i|w_{i-2}, w_{i-1}) = \lambda_1 P_{tri} + \lambda_2 P_{bi} + \lambda_3 P_{uni}$$

Given:

- $\lambda_1 = 0.5$
- $\lambda_2 = 0.3$
- $\lambda_3 = 0.2$

For "data science is":

Trigram probability

$$P(is|data, science) = C(data\ science)C(data\ science\ is)$$

Given $C(data\ science\ is) = 1$ and $C(data\ science) = 3$:

$$P_{tri} = \frac{1}{3} = 0.3333$$

Bigram probability

$$P(is|science) = C(science)C(science\ is)$$

There are 3 occurrences of "science" and 2 are followed by "is":

$$P_{bi} = \frac{2}{3} = 0.6667$$

Unigram probability

The corpus contains 12 words in total, and "is" occurs twice.

$$P(is) = \frac{2}{12} = \frac{1}{6} = 0.1667$$

Now:

$$P(\text{is}|\text{data},\text{science})=(0.5)(0.3333)+(0.3)(0.6667)+(0.2)(0.1667) =0.1667+0.2000+0.0333$$

$$P(\text{is}|\text{data},\text{science})=0.40$$

Interpretation:

Deleted interpolation combines:

- **Trigram:** more context-specific
- **Bigram:** more reliable when trigram data is sparse
- **Unigram:** provides general word-level coverage

Therefore, interpolation provides a good balance between **accuracy and coverage**, especially when training data is small.

4. Entropy of $P(\text{is})=0.66$ and $P(\text{drives})=0.33$

Entropy is:

$$H=-\sum P(x)\log_2 P(x)$$

Therefore:

$$H=-(0.66\log_2 0.66+0.33\log_2 0.33) \quad H\approx-(0.66(-0.599)+0.33(-1.599)) \quad H\approx 0.395+0.528 \quad H\approx 0.923 \text{ bits}$$

The given probabilities add to **0.99** because of rounding. If normalized to exactly 1, the entropy is approximately **0.918 bits**.

Interpretation:

Entropy measures the **uncertainty** of the prediction.

- Low entropy → one prediction is strongly favored.
- High entropy → several predictions are similarly likely.

Here, **"is" (0.66)** is more likely than **"drives" (0.33)**, but there is still uncertainty.

In a keyboard system, entropy can help determine **prediction confidence**. Lower entropy means the keyboard can confidently suggest the top word.

CASE STUDY 2: AI-Powered Customer Support Chatbot

6. Penn Treebank POS Tags

Sentence 1:

"Book a flight ticket now."

Word	POS Tag	Meaning
Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Book/VB a/DT flight/NN ticket/NN now/RB

Here, **"Book" is a verb** because it represents an action/request

Sentence 2:

"This book is interesting."

Word	POS Tag	Meaning
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb

interesting	JJ	Adjective
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This/DT book/NN is/VBZ interesting/JJ

Here, **"book"** is a **noun** because it refers to an object.

Context is important:

The same word **"book"** can have different grammatical roles:

- **Book a flight** → Verb
- **This book** → Noun

7. HMM likelihood of "book" being VB

For an HMM:

$$P(\text{VB}, \text{book}) = P(\text{VB} | \text{Start}) \times P(\text{book} | \text{VB})$$

Given:

$$P(\text{VB} | \text{Start}) = 0.5 \quad P(\text{book} | \text{VB}) = 0.6$$

Therefore:

$$P(\text{VB}, \text{book}) = 0.5 \times 0.6 \quad P(\text{VB}, \text{book}) = 0.30$$

For comparison, noun:

$$P(\text{NN}, \text{book}) = P(\text{NN} | \text{Start}) \times P(\text{book} | \text{NN}) = 0.5 \times 0.4 = 0.20$$

Therefore:

$$0.30 > 0.20$$

So the HMM favors **VB**.

Reason:

The word "book" has a higher emission probability as a verb:

$$P(\text{book}|\text{VB})=0.6$$

compared with:

$$P(\text{book}|\text{NN})=0.4$$

Hence, the probabilistic model favors **VB** in this context.

8. Rule-Based Tagging vs HMM Tagging

Feature	Rule-Based Tagging	HMM/Stochastic Tagging
Method	Uses manually written rules	Uses probabilities
Context	Uses predefined grammatical rules	Learns patterns from corpus
Ambiguity	Can resolve using rules	Resolves using probabilities
Training	Usually requires manual rule creation	Requires tagged training data
Scalability	Difficult for large systems	Better for large datasets
Adaptability	Less flexible	More flexible

Recommendation

For a large-scale chatbot, HMM/Stochastic tagging is more suitable.

Reason: HMM can learn from large amounts of language data and assign the most probable tag based on context. It is better suited for handling ambiguous words such as:

- book → VB / NN
- can → MD / NN
- light → NN / JJ / VB

A modern chatbot may use more advanced neural POS models today, but between the two approaches given in the question, **HMM is preferable**.

9. Role of Standardized POS Tagsets and Word Classes

A standardized POS tagset such as the **Penn Treebank tagset** gives words consistent grammatical labels.

For example:

- NN → Noun
- VB → Verb
- JJ → Adjective
- RB → Adverb
- DT → Determiner

This improves chatbot performance in several ways:

Intent Detection

POS information helps identify whether the user is requesting an action, asking about an object, or describing something.

Response Generation

The chatbot can understand the grammatical structure and generate more appropriate responses.

Ambiguity Resolution

Words such as "**book**" can be interpreted correctly based on context.

Better NLP Processing

POS tags help with parsing, information extraction, sentiment analysis, and question answering.

Conclusion:

Standardized POS tags provide a common linguistic representation that improves intent detection, response generation, ambiguity resolution, and overall chatbot accuracy.

CASE STUDY 3: News Analytics and POS Tag Correction System

10. Apply the Transformation Rule

Initial tags:

economic/JJ growth/NN increases/NNS employment/NN

Rule:

Change NNS to VBZ if the preceding word is tagged as NN.

The word "**increases**" is tagged NNS, and the preceding word "**growth**" is tagged NN.

Therefore, apply the rule:

increases/NNS → increases/VBZ

Corrected sequence:

economic/JJ growth/NN increases/VBZ employment/NN

Why it is appropriate:

"Growth" is the subject, and "increases" is the singular present-tense verb describing what growth does.

For example:

Economic growth increases employment.

Here:

- Subject = economic growth
- Verb = increases
- Object = employment

So **VBZ** is correct.

11. Analyze Initial Tagging Errors

The initial tagging was:

economic/JJ growth/NN increases/NNS employment/NN

economic/JJ

Correct because "economic" describes the noun "growth".

growth/NN

Correct because "growth" is a noun and functions as the subject.

increases/NNS

Incorrect because NNS represents a **plural noun**.

The word "increases" is functioning as a **verb**, not a plural noun.

Therefore:

NNS→VBZ

employment/NN

Correct because "employment" is a noun and functions as the object.

Final tags:

economic/JJ growth/NN increases/VBZ employment/NN

The transformation rule successfully fixes the grammatical error.

12. Word Frequency Distribution

Given:

Word	Frequency
economic	120
growth	450
increases	210
employment	380

Total frequency:

$$120+450+210+380=1160$$

The probability/frequency distribution is:

$$P(\text{word}) = \frac{\text{Total Frequency}}{\text{Frequency}(\text{word})}$$

Word	Frequency	Probability	Percentage
economic	120	0.1034	10.34%
growth	450	0.3879	38.79%
increases	210	0.1810	18.10%
employment	380	0.3276	32.76%
Total	1160	1.0000	100%

Interpretation:

- "growth" is the most frequent word.
- "economic" is the least frequent among these four.

- Frequency information helps an NLP system estimate how likely words and tags are.

For example, if a word frequently occurs as a particular POS tag, the tagger can assign a higher probability to that tag.

13. Transformation-Based Tagging and Entropy

Improvement in Tagging Accuracy

The initial tagger produced:

economic/JJ growth/NN increases/NNS employment/NN

There is an error in the tag of "**increases**".

The transformation-based tagger applies:

NNS→VBZ

and produces:

economic/JJ growth/NN increases/VBZ employment/NN

Therefore, the transformation-based tagger improves accuracy by using **learned linguistic rules** to correct systematic tagging errors.

Role of Entropy

Entropy measures uncertainty:

$$H = -\sum P(t_i) \log_2 P(t_i)$$

where t_i represents possible POS tags.

Suppose the word "**increases**" initially has uncertainty between NNS and VBZ:

$$P(\text{NNS})=0.5, P(\text{VBZ})=0.5$$

Then:

$$H=-(0.5\log_2 0.5 + 0.5\log_2 0.5) \quad H=1 \text{ bit}$$

This represents high uncertainty.

After applying the transformation rule, suppose the system becomes highly confident that the tag is VBZ:

$$P(\text{VBZ}) \approx 1$$

Then:

$$H \approx 0$$

So uncertainty decreases.

Overall conclusion

Transformation-Based Tagging:

- Corrects systematic POS errors.
- Uses contextual rules.
- Improves grammatical consistency.
- Reduces ambiguity.
- Can increase tagging accuracy.

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
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	Understa nding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
	Applicati on of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
	Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
	Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
	Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		
	Q. No. / Criteri a Level	Understandin g of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
	1							
	2							
	3							
	Faculty In-charge		Course Coordinator		Program Director			
	Signature: _____		Signature: _____		Signature: _____			
	Date: _____		Date: _____		Date: _____			